AI SYSTEM BRIEF & ARCHITECTURE SPECIFICATION

Automated Lead Qualification & Routing Workflow

Platform: n8n Orchestration | **Pattern:** Two-Stage LLM Extraction & Priority Routing | **Repository:** `lead-routing-automation/`

1. Executive Overview

The **Automated Lead Qualification & Routing Workflow** is a real-time commercial intake pipeline built in n8n and powered by Google Gemini. It captures inbound business enquiries via webhook, extracts structured firmographic and intent attributes, scores commercial priority across four tiers (`high`, `medium`, `low`, `spam`), and deterministically routes each submission to real-time alerts or structured CRM queues.

2. Core Business Problem

> **Operational Bottleneck:** Sales and partnership teams frequently spend hours manually reviewing high volumes of inbound contact forms—where routine small enquiries and spam obscure high-value strategic opportunities, delaying response times on critical commercial leads.

3. Architecture & Execution Flow

Stage	n8n Node / Component	Functional Responsibility
1. Webhook Ingestion	`Webhook` (POST)	Receives structured JSON form payloads from inbound web or partnership contact endpoints.
2. Entity Extraction	`Message a model` (Gemini) → `parse extract`	Extracts company size, enquiry category, and risk/spam indicators into a parsed JSON object.
3. Priority Scoring	`classify2` (Gemini) → `parse classify`	Evaluates extracted signals against business rules to assign a strict priority tier (`high`, `medium`, `low`, or `spam`).
4. Rule-Based Routing	`Switch` (Mode: Rules)	Evaluates top-level priority classification and branches execution across four distinct operational paths.
5. Priority Dispatch	`Send a message` (Slack) & `Create a record` (Airtable)	Triggers instant Slack alerts for high-priority opportunities while logging nurture, archive, and spam records into separate database queues.

4. Key Architectural Design Principles

- **Decoupled Extraction & Classification:** Separating unstructured data extraction from priority scoring improves classification accuracy, simplifies prompt maintenance, and creates auditable intermediate objects.
- **Explicit JSON Unwrapping:** Dedicated parsing nodes (`parse extract` and `parse classify`) normalize raw model responses into clean top-level fields prior to conditional switching.
- **Noise Suppression & SLA Acceleration:** Isolates spam and low-fit submissions into silent database archives while guaranteeing immediate team visibility for high-value commercial leads.